Fermi–Amaldi-Based Local Mixing Function

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I. MAIN TEXT

A local hybrid exchange functional is given by

$$E_{\mathrm{x}}^{\mathrm{LH}} = \int \left\{ g(\mathbf{r}) e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) + [1 - g(\mathbf{r})] e_{\mathrm{x}}^{\mathrm{SL}}(\mathbf{r}) \right\} d\mathbf{r}, \qquad (1)$$

where the integrand is a mixture of two exchange density energies weighed by a local mixing function (LMF), $g(\mathbf{r})$, which is position-dependent. The $e_{\mathrm{x}}^{\mathrm{SL}}$ and $e_{\mathrm{x}}^{\mathrm{exact}}$ quantities are a semi-local and the exact exchange energy densities, respectively. The exact exchange density (closed-shell) is given by

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r};\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \qquad (2)$$

where $\gamma(\mathbf{r};\mathbf{r}') = 2\sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r})\varphi_i(\mathbf{r}')$ is one-electron reduced density matrix [1, 2].

Here, we propose a new local mixing function for local hybrid functionals. This LMF is the ratio between the Fermi–Amaldi and the exact exchange energy densities:

$$g(\mathbf{r}) = \frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}}. \qquad (3)$$

The Fermi–Amaldi [3, 4] energy density is expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N,

$$e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \qquad (4)$$

Both exchange densities are manifestly non-positive and equivalent for one-electron systems and two-electron singlets. Although, one can show that

$$e_{\mathrm{x}}^{\mathrm{FA}} > e_{\mathrm{x}}^{\mathrm{exact}}, \quad N > 2 \qquad (5)$$

since $e_{\mathrm{x}}^{\mathrm{FA}}$ is scaled down by N, therefore,

$$e_{\mathrm{x}}^{\mathrm{exact}} \leq e_{\mathrm{x}}^{\mathrm{FA}} \leq 0. \qquad (6)$$

Combining these properties, we find that the proposed LMF is bound between zero and one:

$$0 \leq \frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} \leq 1. \qquad (7)$$

This inequality ensures that one hundred percent of exact exchange is employed for one-electron systems, satisfying the one-electron constraint [5, 6].

A particular case of a one-electron limit, i.e., the asymptotic limit ($r \to \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r\to\infty} \frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} = 1 \qquad (8)$$

since both energy densities approach $-1/2r$ as $r \to \infty$.

Another important constraint, the high-density limit,

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